

Time Series — Solutions

Practice Exam — Week 12: Non-Linear Time Series — ARCH & GARCH
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Problem 1 (5 points)

(a) Since σ_t is $\mathcal{F}_{t-1}$ -measurable and η_t is independent of $\mathcal{F}_{t-1}$ with $\mathbb{E}[\eta_t] = 0$,

$$\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[\sigma_t \eta_t \mid \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[\eta_t] = \sigma_t \cdot 0 = 0.$$

By the law of iterated (tower) expectation,

$$\mathbb{E}[r_t] = \mathbb{E}[\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]] = \mathbb{E}[0] = 0.$$

The conditional variance is, since $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$ and $\text{Var}(\eta_t) = 1$,

$$\text{Var}(r_t \mid \mathcal{F}_{t-1}) = \mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2 \mathbb{E}[\eta_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2 = \beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2.$$

(b) By the *law of total variance*,

$$\text{Var}(r_t) = \mathbb{E}[\text{Var}(r_t \mid \mathcal{F}_{t-1})] + \text{Var}(\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]) = \mathbb{E}[\sigma_t^2] + \text{Var}(0) = \mathbb{E}[\sigma_t^2].$$

Taking the (unconditional) expectation of the variance recursion and using *linearity of expectation*,

$$\mathbb{E}[\sigma_t^2] = \beta_0 + \beta_1 \mathbb{E}[r_{t-1}^2] + \beta_2 \mathbb{E}[r_{t-2}^2].$$

Because $\mathbb{E}[r_{t-j}] = 0$ we have $\mathbb{E}[r_{t-j}^2] = \text{Var}(r_{t-j})$, and by *stationarity* $\text{Var}(r_t) = \text{Var}(r_{t-1}) = \text{Var}(r_{t-2}) =: v$. Hence

$$v = \beta_0 + \beta_1 v + \beta_2 v \implies (1 - \beta_1 - \beta_2) v = \beta_0 \implies \boxed{\text{Var}(r_t) = \frac{\beta_0}{1 - \beta_1 - \beta_2}} \quad (\beta_1 + \beta_2 < 1).$$

Numerically, with $\beta_0 = 0.2$, $\beta_1 = 0.3$, $\beta_2 = 0.5$:

$$\text{Var}(r_t) = \frac{0.2}{1 - 0.3 - 0.5} = \frac{0.2}{0.2} = 1.$$

(c) For the variance to be positive and finite we need the denominator positive, i.e.

$$\boxed{\beta_1 + \beta_2 < 1} \quad (\text{with } \beta_0 > 0, \beta_1, \beta_2 \geq 0).$$

Here $\beta_1 + \beta_2 = 0.3 + 0.5 = 0.8 < 1$, so a stationary variance exists (and equals 1). This is exactly the $p = 2, q = 0$ case of the general GARCH stationarity rule $\sum_j \alpha_j + \sum_j \beta_j < 1$: an ARCH(2) has α -coefficients β_1, β_2 and no GARCH (β_j) feedback terms, so the rule reads $\beta_1 + \beta_2 < 1$.

Problem 2 (5 points)

(a) *Zero mean.* σ_t is determined by the past, hence $\mathcal{F}_{t-1}$ -measurable, and η_t is independent of $\mathcal{F}_{t-1}$ with $\mathbb{E}[\eta_t] = 0$. By the tower property,

$$\mathbb{E}[r_t] = \mathbb{E}[\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]] = \mathbb{E}[\sigma_t \mathbb{E}[\eta_t \mid \mathcal{F}_{t-1}]] = \mathbb{E}[\sigma_t \cdot 0] = 0.$$

Uncorrelated. Fix $\tau > 0$. Since $\mathbb{E}[r_t] = 0$, $\text{Cov}(r_{t+\tau}, r_t) = \mathbb{E}[r_{t+\tau} r_t]$. Both r_t and $\sigma_{t+\tau}$ are $\mathcal{F}_{t+\tau-1}$ -measurable (they depend only on information up to time $t + \tau - 1$), so conditioning on $\mathcal{F}_{t+\tau-1}$ and pulling them out of the inner expectation,

$$\begin{aligned} \mathbb{E}[r_{t+\tau} r_t] &= \mathbb{E}[\mathbb{E}[\sigma_{t+\tau} \eta_{t+\tau} r_t \mid \mathcal{F}_{t+\tau-1}]] \\ &= \mathbb{E}[r_t \sigma_{t+\tau} \mathbb{E}[\eta_{t+\tau} \mid \mathcal{F}_{t+\tau-1}]] = \mathbb{E}[r_t \sigma_{t+\tau} \cdot 0] = 0, \end{aligned}$$

using $\mathbb{E}[\eta_{t+\tau} | \mathcal{F}_{t+\tau-1}] = \mathbb{E}[\eta_{t+\tau}] = 0$ by independence. By symmetry of the covariance the same holds for $\tau < 0$. Therefore $\text{Cov}(r_{t+\tau}, r_t) = 0$ and $\text{Corr}(r_{t+\tau}, r_t) = 0$ for all $\tau \neq 0$. Combined with $\mathbb{E}[r_t] = 0$ and a constant (stationary) variance, $\{r_t\}$ is **white noise**.

(b) As in Problem 1, $\mathbb{E}[r_{t-j}^2] = \text{Var}(r_{t-j})$ (mean zero) and, using $\mathbb{E}[r_{t-j}^2 | \mathcal{F}_{t-j-1}] = \sigma_{t-j}^2$ together with the tower property, $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2]$; by stationarity all these equal $v := \text{Var}(r_t)$. Taking unconditional expectations of the recursion,

$$\mathbb{E}[\sigma_t^2] = \alpha_0 + \sum_{j=1}^m \alpha_j \mathbb{E}[r_{t-j}^2] + \sum_{j=1}^r \beta_j \mathbb{E}[\sigma_{t-j}^2] \implies v = \alpha_0 + \left(\sum_j \alpha_j + \sum_j \beta_j \right) v.$$

Solving,

$$\boxed{\text{Var}(r_t) = v = \frac{\alpha_0}{1 - \sum_j \alpha_j - \sum_j \beta_j}},$$

which is positive and finite **iff**

$$\boxed{\sum_{j=1}^m \alpha_j + \sum_{j=1}^r \beta_j < 1} \quad (\text{the weak-stationarity condition}).$$

(c) For GARCH(1,1) with $\alpha_0 = 2 \times 10^{-5}$, $\alpha_1 = 0.10$, $\beta_1 = 0.85$:

$$\alpha_1 + \beta_1 = 0.10 + 0.85 = 0.95 < 1,$$

so the process is weakly stationary. Then

$$\text{Var}(r_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1} = \frac{2 \times 10^{-5}}{1 - 0.95} = \frac{2 \times 10^{-5}}{0.05} = 4 \times 10^{-4},$$

and the implied daily volatility is

$$\sqrt{\text{Var}(r_t)} = \sqrt{4 \times 10^{-4}} = 2 \times 10^{-2} = 2\% \text{ per day},$$

a realistic equity figure. The persistence $\alpha_1 + \beta_1 = 0.95$ is close to 1, the hallmark of financial GARCH fits: volatility shocks decay only slowly (sticky volatility, long quiet/turbulent regimes), yet the process stays stationary because the sum is still strictly below 1.

Problem 3 (5 points)

(a) Fix $\tau > 0$ and start from the covariance, using stationarity ($\mathbb{E}[r_{t+\tau}^2] = \mathbb{E}[r_t^2]$):

$$\text{Cov}(r_{t+\tau}^2, r_t^2) = \mathbb{E}[r_{t+\tau}^2 r_t^2] - (\mathbb{E}[r_t^2])^2.$$

Condition on $\mathcal{F}_{t+\tau-1}$: both r_t^2 and $\sigma_{t+\tau}^2 = \alpha_0 + \alpha_1 r_{t+\tau-1}^2$ are measurable there, and $\mathbb{E}[\eta_{t+\tau}^2 | \mathcal{F}_{t+\tau-1}] = \mathbb{E}[\eta_{t+\tau}^2] = 1$, so

$$\mathbb{E}[r_{t+\tau}^2 r_t^2 | \mathcal{F}_{t+\tau-1}] = r_t^2 \sigma_{t+\tau}^2 \mathbb{E}[\eta_{t+\tau}^2 | \mathcal{F}_{t+\tau-1}] = r_t^2 (\alpha_0 + \alpha_1 r_{t+\tau-1}^2).$$

Taking expectations,

$$\mathbb{E}[r_{t+\tau}^2 r_t^2] = \alpha_0 \mathbb{E}[r_t^2] + \alpha_1 \mathbb{E}[r_{t+\tau-1}^2 r_t^2].$$

Write $C_\tau := \text{Cov}(r_{t+\tau}^2, r_t^2)$ and use $\mathbb{E}[r_{t+\tau-1}^2 r_t^2] = C_{\tau-1} + (\mathbb{E}[r_t^2])^2$ (again by stationarity, $\mathbb{E}[r_{t+\tau-1}^2] = \mathbb{E}[r_t^2]$). Then

$$C_\tau + (\mathbb{E}[r_t^2])^2 = \alpha_0 \mathbb{E}[r_t^2] + \alpha_1 (C_{\tau-1} + (\mathbb{E}[r_t^2])^2),$$

i.e.

$$C_\tau = \alpha_1 C_{\tau-1} + \underbrace{\alpha_0 \mathbb{E}[r_t^2] + \alpha_1 (\mathbb{E}[r_t^2])^2 - (\mathbb{E}[r_t^2])^2}_{=:R}.$$

Substitute $\mathbb{E}[r_t^2] = \text{Var}(r_t) = \alpha_0/(1 - \alpha_1)$:

$$R = \alpha_0 \cdot \frac{\alpha_0}{1 - \alpha_1} - (1 - \alpha_1) \left(\frac{\alpha_0}{1 - \alpha_1} \right)^2 = \frac{\alpha_0^2}{1 - \alpha_1} - \frac{\alpha_0^2}{1 - \alpha_1} = 0.$$

Hence

$$\boxed{C_\tau = \alpha_1 C_{\tau-1}} \quad (\tau > 0).$$

Iterating τ times from $C_0 = \text{Var}(r_t^2)$ gives $C_\tau = \alpha_1^\tau \text{Var}(r_t^2)$, so

$$\text{Corr}(r_{t+\tau}^2, r_t^2) = \frac{C_\tau}{\text{Var}(r_t^2)} = \alpha_1^\tau,$$

and by symmetry of the autocovariance, $\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$ for all τ . (The condition $0 < \alpha_1 < 1/\sqrt{3}$, i.e. $\alpha_1^2 < 1/3$, guarantees $\text{Var}(r_t^2) = \mathbb{E}[r_t^4] - (\mathbb{E}[r_t^2])^2 < \infty$, so the correlation is well defined.)

(b) With $\alpha_1 = 0.5$, $\alpha_1^2 = 0.25 < 1/3$, so the fourth moment exists and

$$\kappa = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2} = \frac{3(1 - 0.25)}{1 - 0.75} = \frac{3 \cdot 0.75}{0.25} = \frac{2.25}{0.25} = 9.$$

Since $\kappa = 9 > 3$, the marginal law has *heavier tails than the normal* (excess kurtosis $9 - 3 = 6$): ARCH generates fat tails endogenously, even with Gaussian innovations. The squared-return autocorrelations are $\text{Corr}(r_{t+\tau}^2, r_t^2) = 0.5^{|\tau|}$:

$$\tau = 1 : 0.5, \quad \tau = 2 : 0.25, \quad \tau = 3 : 0.125.$$

The two thresholds differ because they control different moments: $\alpha_1 < 1$ keeps the *second* moment ($\text{Var } r_t = \alpha_0/(1 - \alpha_1)$) finite, whereas the *fourth* moment involves $\mathbb{E}[\sigma_t^4]$ and the recursion for it carries the factor $3\alpha_1^2$ (from $\mathbb{E}[\eta^4] = 3$); a finite solution needs $3\alpha_1^2 < 1$, i.e. the strictly tighter $\alpha_1^2 < 1/3$. As $\alpha_1^2 \uparrow 1/3$ the kurtosis $\rightarrow \infty$.

(c) With $\alpha_0 = 4 \times 10^{-4}$, $\alpha_1 = 0.5$ and $r_t = 0.04$ (so $r_t^2 = 1.6 \times 10^{-3}$),

$$\sigma_{t+1}^2 = \alpha_0 + \alpha_1 r_t^2 = 4 \times 10^{-4} + 0.5 (1.6 \times 10^{-3}) = 4 \times 10^{-4} + 8 \times 10^{-4} = 1.2 \times 10^{-3},$$

so $\sigma_{t+1} = \sqrt{1.2 \times 10^{-3}} \approx 0.0346$. Since $r_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(0, \sigma_{t+1}^2)$, the 95% conditional predictive interval is

$$0 \pm 1.96 \sigma_{t+1} = \pm 1.96 (0.0346) \approx \pm 0.0679.$$

The unconditional variance is $\text{Var}(r_t) = \alpha_0/(1 - \alpha_1) = 4 \times 10^{-4}/0.5 = 8 \times 10^{-4}$, so $\sqrt{\text{Var}(r_t)} \approx 0.0283$ and the unconditional band is $\pm 1.96 (0.0283) \approx \pm 0.0554$. The conditional band (± 0.068) is markedly *wider* than the unconditional one (± 0.055): after the large move $r_t = 0.04$ the model raises its volatility forecast — exactly the volatility-clustering behaviour ARCH is designed to capture.

Problem 4 (5 points)

(a) With $n = 100$, $n(n+2) = 100 \cdot 102 = 10200$, and $m = 5$:

$$Q_5 = 10200 \left(\frac{0.05^2}{99} + \frac{(-0.12)^2}{98} + \frac{0.08^2}{97} + \frac{(-0.10)^2}{96} + \frac{0.06^2}{95} \right).$$

Term by term,

$$\begin{aligned}\frac{0.0025}{99} &= 2.525 \times 10^{-5}, & \frac{0.0144}{98} &= 1.469 \times 10^{-4}, & \frac{0.0064}{97} &= 6.598 \times 10^{-5}, \\ \frac{0.0100}{96} &= 1.042 \times 10^{-4}, & \frac{0.0036}{95} &= 3.789 \times 10^{-5}.\end{aligned}$$

Summing the five terms gives 3.802×10^{-4} , so

$$Q_5 \approx 10200 \times 3.802 \times 10^{-4} \approx \boxed{3.88}.$$

(b) For an ARMA(1, 1), $p = q = 1$, so $\text{df} = m - p - q = 5 - 1 - 1 = 3$. The 5% critical value is $\chi_{3,0.95}^2 = 7.815$. Since $Q_5 \approx 3.88 < 7.815$, we **do not reject** H_0 : there is no significant evidence of residual autocorrelation up to lag 5, so the ARMA(1, 1) appears *adequate*.

Had an AR(1) been fitted, $p = 1, q = 0$, so $\text{df} = 5 - 1 - 0 = 4$ and the critical value rises to $\chi_{4,0.95}^2 = 9.488$. Since $3.88 < 9.488$, the decision is unchanged (do not reject). In general fewer fitted parameters leave more degrees of freedom and widen the acceptance region.

(c) The parameter counts are $k = p + q + 1$: for ARMA(1, 1), $k = 3$; for AR(3), $k = 4$ (one extra parameter). With $n = 100$, $\log n = \log 100 \approx 4.605$. Comparing penalties when adding the one extra parameter:

$$\begin{aligned}\Delta(\text{penalty})_{\text{AIC}} &= 2(k_{\text{big}} - k_{\text{small}}) = 2, \\ \Delta(\text{penalty})_{\text{BIC}} &= (\log n)(k_{\text{big}} - k_{\text{small}}) = \log 100 \approx 4.605.\end{aligned}$$

So the AR(3) is preferred only if $-2 \log L$ drops by more than the extra penalty:

$$\text{AIC: drop} > 2 \text{ (i.e. } \log L \text{ improves by } > 1); \quad \text{BIC: drop} > 4.605 \text{ (} \log L \text{ improves by } > 2.30).$$

Since BIC's $\log n = 4.605 > 2$ charges more per parameter (as $n > e^2 \approx 7.4$), **AIC is more willing to select the larger AR(3) model**; BIC is harder to satisfy and favours the parsimonious ARMA(1, 1).

In-sample $R^2 = 1 - s_e^2/s_y^2$ is a poor selection tool because adding parameters can only *increase* it (the larger nested model fits the training data at least as well), so a higher R^2 for AR(3) is no evidence of a genuinely better model; moreover even the true model has a bounded R^2 (a stochastic process is not perfectly predictable). Information criteria penalise complexity and are the correct basis for the comparison.